

LE MARS, IA, USA

WMO: 725484

Lat: 42.778N

Lon: 96.194W

Elev: 1196

StdP: 14.07

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04942

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -7.7	(c) -2.1	(d) -14.7	(e) 2.6	(f) -5.8	(g) -9.2	(h) 3.5	(i) -1.2	(j) 28.1	(k) 20.6	(l) 25.6	(m) 22.3	(n) 4.1	(o) 320	(p) 0.562

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.6	92.7	76.4	90.2	75.2	87.7	73.5	80.1	88.6	78.0	86.0	75.9	84.2	12.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.4	149.0	85.9	75.2	138.2	84.0	73.0	128.0	81.7	44.6	88.0	42.5	85.9	40.3	85.0	93.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
25.1	21.9	19.4	-16.4	96.7	5.8	3.0	-20.6	98.8	-24.0	100.6	-27.2	102.2	-31.4	104.4		
			-17.1	82.5	5.5	2.5	-21.1	84.3	-24.3	85.7	-27.5	87.1	-31.5	88.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	21.1	24.2	36.9	49.0	60.2	71.2	74.7	72.2	64.2	51.4	36.6	24.6
	DBStd	21.41	12.45	12.72	12.20	10.30	8.67	6.73	6.20	5.86	8.92	9.27	10.79	11.12
	HDD50	3508	896	723	431	138	14	0	0	0	6	96	417	787
	HDD65	6755	1361	1142	871	487	199	20	6	9	123	432	853	1252
	CDD50	3137	0	0	25	108	332	637	765	687	432	138	13	0
	CDD65	910	0	0	1	7	51	207	306	231	98	9	0	0
	CDH74	9403	0	0	17	133	618	2165	3258	2116	969	124	3	0
	CDH80	3402	0	0	1	36	194	815	1304	709	314	29	0	0
	WSAvg	8.3	8.5	8.8	9.5	10.4	9.6	8.1	6.4	6.1	7.3	8.1	8.5	8.2
	PrecAvg	24.0	0.5	0.6	1.7	2.5	3.3	3.7	2.8	3.1	2.8	1.7	1.0	0.6
(15) Precipitation	PrecMax	36.5	2.0	1.7	4.8	7.5	8.9	8.0	8.4	12.5	8.9	5.1	4.0	2.3
	PrecMin	11.1	0.0	0.0	0.0	0.4	0.5	0.7	0.5	0.4	0.6	0.0	0.0	0.0
	PrecStd	5.4	0.4	0.4	1.2	1.6	1.4	2.0	1.7	2.2	1.8	1.3	0.9	0.5
	PrecStd	5.4	0.4	0.4	1.2	1.6	1.4	2.0	1.7	2.2	1.8	1.3	0.9	0.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	55.0	59.2	76.8	84.4	90.5	95.2	97.0	94.8	92.5	84.0	71.8	54.9	47.5
	0.4% MCWB	43.9	49.5	57.6	62.2	67.7	75.6	78.5	79.0	73.0	64.9	56.0	47.5	47.5
	2% DB	45.7	50.4	68.4	77.2	85.5	90.9	92.8	90.4	87.8	77.2	63.6	48.4	42.7
	2% MCWB	39.4	43.8	54.8	59.2	66.9	74.1	77.9	77.8	71.5	60.1	52.5	42.7	42.7
	5% DB	40.8	45.1	61.4	72.3	81.3	88.2	90.2	86.6	82.9	72.5	57.4	43.3	38.0
	5% MCWB	36.4	40.2	52.5	56.8	63.7	72.0	76.9	75.7	69.5	58.7	47.6	38.0	38.0
	10% DB	36.8	40.6	54.9	67.5	76.7	84.3	87.5	83.8	80.6	68.0	52.4	38.9	35.6
	10% MCWB	34.3	37.0	47.2	54.3	61.6	70.1	75.3	74.2	68.1	56.7	45.2	35.6	35.6
	0.4% WB	44.7	51.9	62.5	66.2	74.3	80.1	85.2	83.0	77.2	69.8	58.9	50.0	50.0
	0.4% MCDB	53.7	60.9	70.4	76.5	83.8	90.1	92.7	91.3	86.0	77.1	66.0	53.2	53.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	40.1	44.5	57.2	61.5	69.8	77.5	81.0	79.5	74.9	64.3	53.4	42.5	42.5
	2% MCDB	45.2	49.3	65.0	73.9	80.4	87.1	89.3	88.0	83.2	73.2	63.1	47.7	47.7
	5% WB	37.1	40.3	52.9	58.6	66.6	75.1	78.6	77.3	72.8	61.0	49.4	38.9	38.9
	5% MCDB	40.2	44.7	61.1	69.6	76.6	84.9	86.4	84.6	80.3	69.2	57.2	43.4	43.4
	10% WB	34.4	37.2	47.6	55.4	63.9	72.5	76.6	75.0	70.2	57.8	45.7	36.0	36.0
	10% MCDB	37.2	40.5	56.1	65.2	73.1	81.8	84.4	82.3	77.5	65.7	51.2	39.0	39.0
(35) Mean Daily Temperature Range	MDBR	18.0	17.9	19.7	22.6	21.8	20.8	20.6	20.1	23.4	22.9	20.2	16.9	16.9
	5% DB	23.6	23.2	31.4	31.2	28.4	23.6	22.0	22.3	26.3	30.6	28.9	23.1	23.1
	5% WB	17.9	17.3	19.3	16.3	13.4	11.1	11.6	12.8	13.9	16.9	18.2	17.6	17.6
	MCDBR	22.4	21.7	29.0	25.9	23.2	20.3	19.3	19.6	21.5	24.7	26.6	21.9	21.9
	MCWBR	17.3	16.5	18.4	15.3	13.3	12.1	13.0	12.5	13.0	16.0	17.8	17.3	17.3
	MCWBR	17.3	16.5	18.4	15.3	13.3	12.1	13.0	12.5	13.0	16.0	17.8	17.3	17.3
(40) Clear-Sky Solar Irradiance	taub	0.277	0.287	0.322	0.378	0.405	0.410	0.420	0.416	0.369	0.330	0.294	0.281	0.281
	taud	2.425	2.420	2.405	2.264	2.264	2.311	2.278	2.278	2.380	2.468	2.490	2.435	2.435
	Ebn at Noon	276	292	293	281	276	274	270	267	273	272	268	263	263
	Edh at Noon	24	28	33	41	42	40	41	40	34	27	23	22	22
(44) All-Sky Solar Radiation	RadAvg	609	950	1217	1542	1797	2008	2098	1752	1413	971	674	495	495
	RadStd	55	76	95	127	132	146	100	127	95	119	43	40	40

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (47 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page